

So I hear you'd like
to do some ranked
coding...

Introduction to Competitive Programming

Workshop Agenda

- Who is KCPC?
- What does KCPC do?
- Why start KCPC?
- What is Competitive Programming?
- Why do Competitive Programming?
- Some nice problems
- Final messages

But first attendance...

→ Attendance



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- Partnered with Orthant Research Group under the Orthant Research Europe division

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- Build a cross-University collaborative network

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But creating them to be fast and small is where most of the challenges lie.

Requirements for designing algorithms

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Don't worry if this seems too much, you only need GCSE level mathematics to get started!

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You won't need to make the code maintainable like in software engineering.

But most solutions are at most a few hundred lines of codes anywhere so making your solution concise and straightforward.

Example of a Competitive Programming problem

→ Ex: Problem description

A. Team

time limit per test: 2 seconds

memory limit per test: 256 megabytes

One day three best friends Petya, Vasya and Tonya decided to form a team and take part in programming contests. Participants are usually offered several problems during programming contests. Long before the start the friends decided that they will implement a problem if at least two of them are sure about the solution. Otherwise, the friends won't write the problem's solution.

This contest offers n problems to the participants. For each problem we know, which friend is sure about the solution. Help the friends find the number of problems for which they will write a solution.

Input

The first input line contains a single integer n ($1 \leq n \leq 1000$) — the number of problems in the contest. Then n lines contain three integers each, each integer is either 0 or 1. If the first number in the line equals 1, then Petya is sure about the problem's solution, otherwise he isn't sure. The second number shows Vasya's view on the solution, the third number shows Tonya's view. The numbers on the lines are separated by spaces.

Output

Print a single integer — the number of problems the friends will implement on the contest.

→ Ex: Input and Output

Output

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Examples

input

Copy

```
3
1 1 0
1 1 1
1 0 0
```

output

Copy

```
2
```

input

Copy

```
2
1 0 0
0 1 1
```

output

Copy

```
1
```

→ Ex: Problem notes

Note

In the first sample Petya and Vasya are sure that they know how to solve the first problem and all three of them know how to solve the second problem. That means that they will write solutions for these problems. Only Petya is sure about the solution for the third problem, but that isn't enough, so the friends won't take it.

In the second sample the friends will only implement the second problem, as Vasya and Tonya are sure about the solution.



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- Sharpens your problem-solving skills
- Improves coding accuracy and speed
- Many companies value competitive programming experience
- Deepens understanding of algorithms and data structures
- It's fun

Some nice problems

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→ tinyurl.com/kcpcws

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kcpc.orthant.co.uk/r/Ob464d30-8c8a-4ebe-8270-7c0d868c83c1

Final messages

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- **Too easy today?** Harder challenges are coming
- **Too hard today?** That means you're learning and that's great!
- Either way, **you belong here**: Our goal is simple: **be better than yesterday**
- See you **next week** at the **same time**!